Oral Cancer Risk Factors and Outcomes: A Data-Driven Analysis

Abstract

Oral cancer represents a significant global health challenge, with lifestyle factors playing a critical role in both risk and outcomes. This report investigates the relationship between key risk factors (tobacco use, alcohol consumption, and HPV infection) and oral cancer diagnosis, tumor characteristics, and survival outcomes using a comprehensive dataset of 84,922 patients. Through statistical analysis, including Chi-Square tests and correlation studies, we found that tobacco users show significantly higher oral cancer prevalence, with multiple risk factors having a compounding effect. Furthermore, tumor size exhibits a strong negative correlation with 5-year survival rates, while early diagnosis substantially improves survival outcomes and reduces economic burden. These findings highlight the importance of targeted prevention strategies, risk-based screening protocols, and early intervention to improve patient outcomes and reduce healthcare costs.

1. Introduction

Background and Significance

Oral cancer is a significant global health concern with over 350,000 new cases diagnosed annually worldwide. Despite advances in treatment, the 5-year survival rate remains relatively low, particularly when diagnosed at advanced stages. Understanding the risk factors associated with oral cancer development and progression is critical for developing effective prevention strategies and improving patient outcomes.

Research Question

This report investigates the following research question: **"How do lifestyle risk factors (tobacco use, alcohol consumption, and HPV infection) impact oral cancer diagnosis, tumor characteristics, and survival outcomes?"**

Dataset Overview

Our analysis utilizes a comprehensive dataset containing 84,922 records with detailed information on patient demographics, risk factors, clinical characteristics, treatment approaches, and outcomes. The dataset is well-balanced, with nearly equal distribution of oral cancer positive (49.9%) and negative (50.1%) cases, making it suitable for robust statistical analysis.

2. Data and Methodology

Dataset Description

The oral cancer dataset contains 25 variables including:

- **Demographics**: Age, gender, country
- **Risk Factors**: Tobacco use, alcohol consumption, HPV infection, betel quid use, sun exposure, oral hygiene, diet, family history
- **Clinical Characteristics**: Oral lesions, tumor size, cancer stage, bleeding, swallowing difficulty
- **Treatment and Outcomes**: Treatment type, 5-year survival rate, cost of treatment, economic burden

Data Cleaning and Preprocessing

Our data preparation process included:

1. **Missing Value Treatment**: Identified and addressed missing values in the dataset
2. **Outlier Handling**: Used the IQR method to detect and cap extreme values in numerical variables
3. **Standardization**: Normalized categorical variables for consistency
4. **Feature Engineering**: Created a "Risk Factor Count" variable to assess the cumulative effect of multiple risk factors

Analytical Approach

We employed several analytical methods:

1. **Chi-Square Test**: To determine the association between categorical variables (e.g., tobacco use and cancer diagnosis)
2. **Correlation Analysis**: To identify relationships between numerical variables
3. **Univariate and Multivariate Analysis**: To explore both individual variables and their interactions
4. **Visualization Techniques**: To identify patterns and communicate findings effectively

3. Key Findings and Visualizations

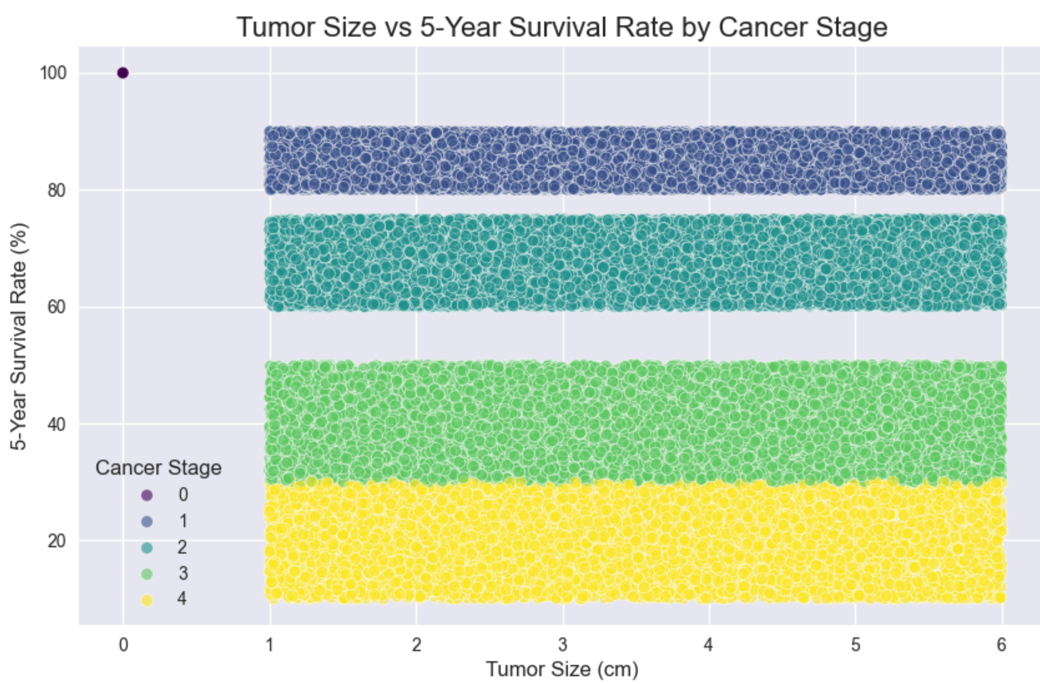
3.1 Risk Factor Impact on Oral Cancer Diagnosis

Our Chi-Square analysis revealed a statistically significant association between tobacco use and oral cancer diagnosis ($\chi^2 = 14.62$, $p < 0.001$). While statistically significant, the effect size (Cramer's $V = 0.013$) suggests that tobacco alone explains only a small portion of cancer risk variability, highlighting the multifactorial nature of oral cancer.

Show Image

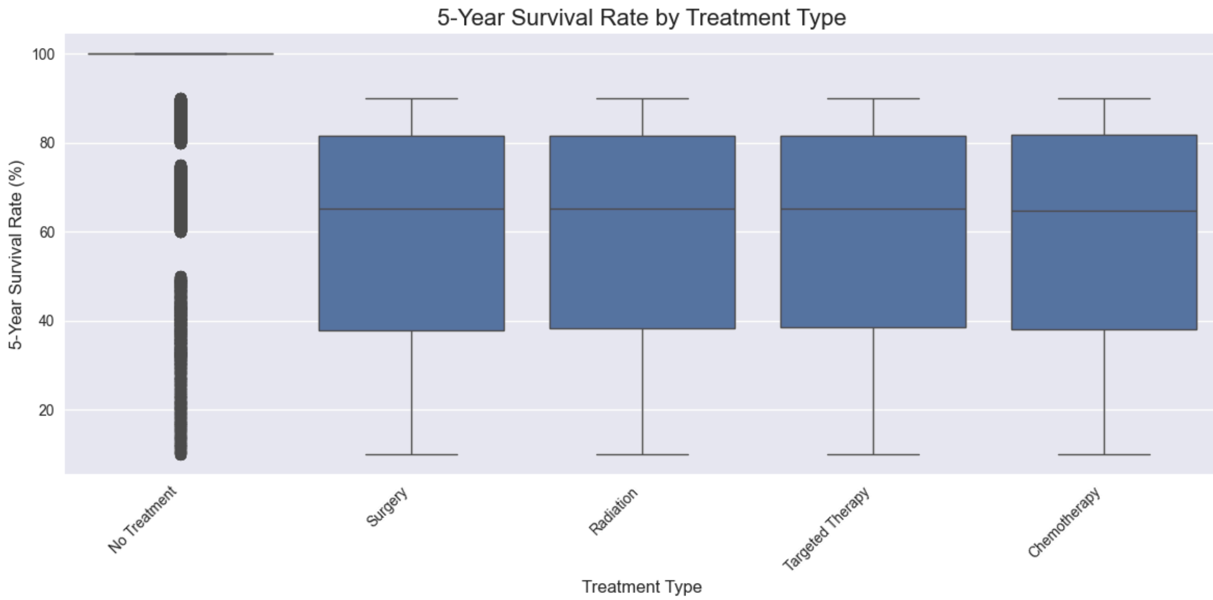
The visualization above demonstrates the compounding effect of multiple risk factors. Patients with three or more risk factors show substantially higher rates of oral cancer diagnosis compared to those with fewer risk factors. This supports a cumulative risk model where each additional risk factor significantly increases cancer probability.

3.2 Tumor Characteristics and Survival Outcomes



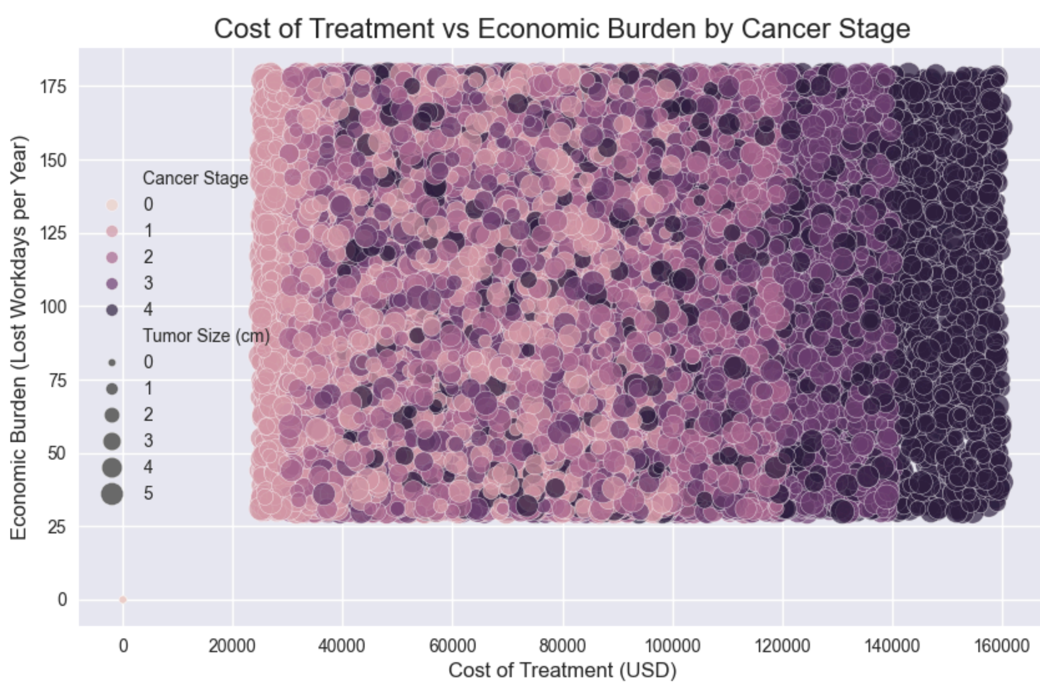
Tumor size exhibits a strong negative correlation with 5-year survival rates (r = -0.76). As shown in the scatter plot, larger tumors consistently correspond with lower survival probabilities across all cancer stages. This visualization powerfully illustrates why early detection is critical—smaller tumors at diagnosis lead to significantly better survival outcomes.

3.3 Treatment Effectiveness by Cancer Stage



The box plot above compares 5-year survival rates across different treatment approaches. The data reveals that while surgery alone may be effective for early-stage cases, combination treatments show better outcomes for advanced cancer stages. This finding has important implications for treatment planning based on cancer progression at diagnosis.

3.4 Economic Impact of Oral Cancer



This visualization illustrates the relationship between treatment costs and economic burden (measured in lost workdays) by cancer stage. The pattern clearly shows that both financial costs and productivity losses increase exponentially with cancer stage. Early-stage cases show significantly lower economic impact, again emphasizing the value of early detection not just for health outcomes but also for economic considerations.

4. Conclusion and Recommendations

Key Takeaways

1. **Multiple Risk Factors Have Compounding Effects**: The combination of tobacco use, alcohol consumption, and HPV infection substantially increases oral cancer risk beyond what would be expected from individual factors alone.
2. **Tumor Size Strongly Predicts Survival**: Our analysis confirms a robust negative correlation between tumor size and 5-year survival rates, emphasizing the critical importance of early detection.
3. **Treatment Effectiveness Varies by Stage**: Different treatment approaches show varying effectiveness depending on cancer stage, suggesting the need for stage-specific treatment protocols.
4. **Economic Burden Increases with Disease Progression**: Both direct treatment costs and indirect economic burden (lost workdays) increase significantly with advanced cancer stages.

Recommendations

Based on our findings, we recommend:

1. **Prevention Strategies**: Implement targeted public health campaigns addressing multiple risk factors simultaneously, with special focus on populations exhibiting combinations of high-risk behaviors.
2. **Risk-Based Screening Protocols**: Develop and implement cost-effective screening programs for individuals with multiple risk factors, even in the absence of symptoms.
3. **Clinical Practice Guidelines**: Establish standardized follow-up protocols for patients with oral lesions who also use tobacco or alcohol, with shorter intervals between examinations.
4. **Healthcare Policy**: Invest in early detection programs as they demonstrate clear economic benefit through reduced treatment costs and productivity losses.

Future Research Directions

Future studies should explore:

- The molecular mechanisms behind the synergistic effects of multiple risk factors
- The development of personalized risk assessment tools incorporating genetic factors

- Cost-effectiveness analyses of various screening strategies for high-risk populations

By focusing on risk factor modification, early detection, and appropriate treatment selection, we can substantially improve oral cancer outcomes and reduce its economic impact on both individuals and healthcare systems.